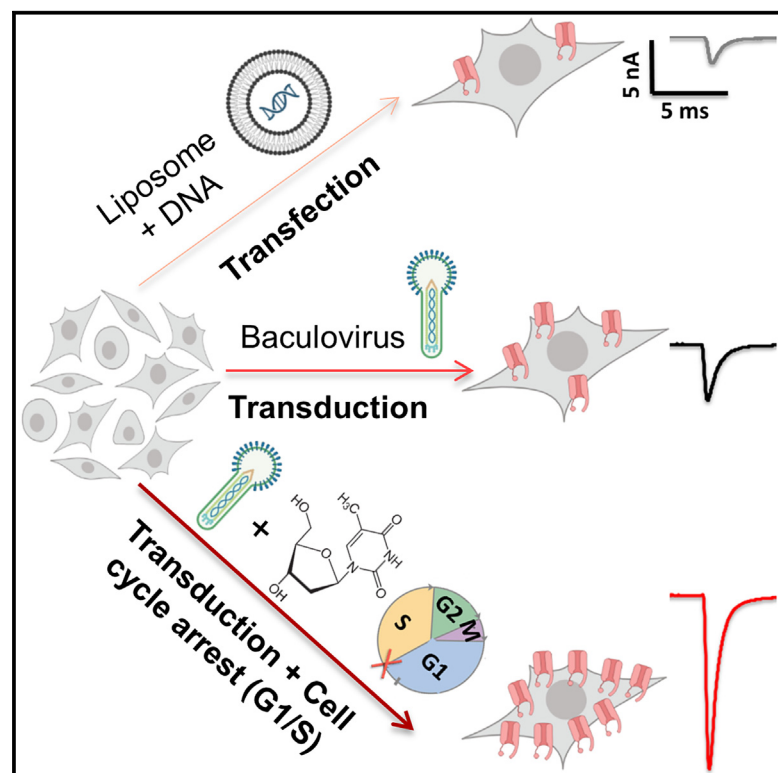


Cell-cycle arrest at the G1/S boundary enhances transient voltage-gated ion channel expression in human and insect cells

Graphical abstract



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In brief

Eltokhi et al. develop a method to improve the heterologous expression of voltage-gated ion channels in human and insect cells by using the baculovirus expression system while arresting the cell cycle at the G1/S boundary. This modified method of expression allows a ~ 7 -fold increase in the peak current density of mammalian sodium channels compared with the current literature.

Highlights

- Baculovirus and cell-cycle arrest enhance expression of voltage-gated ion channels
- Expression in human cells is increased >7 -fold at the G1/S boundary
- Arresting cell cycle in insect cells enhances transient expression by 3-fold
- This method facilitates biophysical characterization of loss-of-function mutants



Article

Cell-cycle arrest at the G1/S boundary enhances transient voltage-gated ion channel expression in human and insect cells

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MOTIVATION Voltage-gated ion channels are among the most challenging proteins to study in heterologous expression systems owing to their low expression and complex structure. Biophysical analysis of those channels requires a high-level expression to allow recording of small ionic currents through the central pore and detection of much smaller gating currents that arise from the transmembrane movement of the voltage-sensing domain. We therefore developed a technique that combines a baculovirus expression system and cell-cycle arrest at the G1/S boundary using either thymidine or hydroxyurea to achieve a multi-fold increase in ion channel expression compared with the classical liposome-mediated transfection methods.

SUMMARY

Heterologous expression of recombinant ion channel subunits in cell lines is often limited by the presence of a low number of channels at the cell surface level. Here, we introduce a combination of two techniques: viral expression using the baculovirus system plus cell-cycle arrest at the G1/S boundary using either thymidine or hydroxyurea. This method achieved a manifold increase in the peak current density of expressed ion channels compared with the classical liposome-mediated transfection methods. The enhanced ionic current was accompanied by an increase in the density of gating charges, confirming that the increased yield of protein and ionic current reflects the functional localization of channels in the plasma membrane. This modified method of viral expression coordinated with the cell cycle arrest will pave the way to better decipher the structure and function of ion channels and their association with ion channelopathies.

INTRODUCTION

Mammalian voltage-gated ion channels including sodium (Na_v), potassium (K_v), and calcium (Ca_v) are among the most challenging proteins to study in heterologous expression systems owing to their low expression and complex structure. Na_v channels are composed of 24 transmembrane segments with large connecting linkers and a central pore.^{1,2} They initiate action potentials by conducting Na^+ into the cell rapidly and selectively.³ Biophysical analysis of Na_v channels requires a high-level expression to allow recording of small ionic currents through the central pore and detection of much smaller gating currents that arise from the transmembrane movement of the voltage-dependent gating apparatus of the channel. On the other hand, K_v channels are tetramers of four identical subunits, each composed of six membrane segments, arranged as a ring and contributing to the wall of the transmembrane K^+ -selective pore.⁴ They are vital in repolarizing the membrane

to a negative resting potential to terminate the action potential signal.⁵ Although expressing K_v channels in heterologous models is easier than in Na_v channels, there can still be challenges associated with ensuring proper folding, selecting an appropriate expression system, and obtaining functional channels at high levels required for studying their structures and functions.

Many missense variants that target ion channels lead to serious diseases, which are caused by altered function, reduced expression, or defects in trafficking to the cell membrane, adding a second level of complexity in studying those channels.^{6,7} The pathogenic dysfunction of ion channels caused by some missense mutations leads to small, but pathophysiologically significant, changes in voltage dependence and ion selectivity. Moreover, some missense mutations in the Na_v and K_v channels target the voltage sensor and affect the small capacitive gating currents generated by the outward movement of the gating charges, thus changing sensitivity to



membrane potential or inducing even smaller ionic leaks through the voltage sensor that produce pathogenic gating pore current. Gating currents and gating pore currents are in the range of 0.1%–1% of the peak sodium current,^{8,9} requiring high levels of expression and sensitive biophysical measurements to analyze them. Measuring such important biophysical features is hindered by the typically low expression of Na_v and certain K_v channels in mammalian cell lines. Classical heterologous expression of recombinant DNA via liposome-mediated transfection, as a means of introducing genetic material, is an important method for investigating protein function, with several benefits including efficient transcription, faithful translation, proper protein folding, and post-translational modifications.¹⁰ However, the transcription level of recombinant DNA in these cellular models does not often correlate with the expression of high-quality, fully functional, biologically active proteins.¹¹ Moreover, the expression of several classes of complex membrane proteins, such as voltage-gated ion channels, often results in low protein yield and low copy numbers of protein localized on the surface of a cell, which are insufficient to support detailed biophysical analyses that are needed for structure-function studies and the characterization of disease mutations.¹²

Here, we have addressed the problem of expressing complex membrane proteins with two synergistic approaches: high-level viral expression using methods developed for structural biology plus cell-cycle arrest to allow maximum transcription and translation. DNA is introduced into the heterologous expression system using the baculovirus vector system, which is widely used for the expression of high levels of recombinant proteins in cultured insect cells, such as Hi-5 and Sf9 cells isolated from the moths *Trichoplusia ni* and *Spodoptera frugiperda*, respectively. This expression system is widely used in the large-scale preparation of proteins for analysis of their structure by X-ray crystallography and cryoelectron microscopy (cryo-EM),^{13–15} Baculovirus genes are not expressed in mammalian cells; however, inserting a promoter recognized by mammalian transcription factors allows transcription of mammalian genes, as in the BacMam system.^{16–18} This system uses a modified baculovirus vector as a vehicle to efficiently express genes in mammalian cells. Use of the original baculovirus alone or with the modified BacMam infection system leads to a high level of expression of different types of Na_v channels for structure-function studies,^{19,20} as well as the ability to detect small electrical signals generated by gating currents and gating pore currents.^{21,22}

In the method presented here, we have combined the BacMam expression system with cell-cycle arrest to synchronize and prolong the period of productive protein expression in mammalian cell lines during and following viral transduction. Performing cell-cycle arrest at the G1/S boundary using either thymidine or hydroxyurea before and during the transduction of recombinant baculoviruses containing mammalian cell-active expression cassettes (e.g., the BacMam system)^{16,23} with the powerful cytomegalovirus (CMV) promoter showed a substantial increase in the fluorescence signal of the GFP reporter and a corresponding increase in the peak current density and gating charge movement of mammalian voltage-gated sodium channels in two mammalian cell lines, HEK293 and tsA201. The cell-cycle arrest, in combina-

tion with the original baculovirus system, was also applicable in enhancing the expression of mammalian and bacterial K_v7 channels in two insect cell lines, Hi-5 and Sf9, highlighting the cell-cycle arrest at the G1/S boundary to be a versatile tool for increasing the transient expression of ion channels and, possibly, other complex proteins in different cell lines.

RESULTS

Effects of cell-cycle arrest at M and G1 phases on sodium channel expression

We began by testing whether blocking the growth of the human embryonic kidney (HEK) cells before and during the viral transduction would enhance the expression of Na_v channels and the reporter GFP protein. Several compounds are known to induce cell-cycle arrest at different stages (Figure 1A). Therefore, we tested the effects of treating HEK293 cells with either nocodazole or lovastatin, which arrest cell growth at the M and G1 cycles, respectively. We added each compound individually, 24 h before viral transduction of human $\text{Na}_v1.2$ channels, and the same drug concentration was maintained after the viral infection phase. After 48 h, we measured the fluorescence signal of the GFP reporter protein (Figures 1B, and S1A). Arresting the growth of HEK293 cells at the M phase of the cell cycle using nocodazole revealed a 3-fold increase in the GFP fluorescence signal (Figure S1A). In contrast, arresting HEK293 cells at the G1 cycle using lovastatin showed a trend toward decreasing the GFP fluorescence signals, which did not reach significance (Figure S1A). Building on the promising fluorescence results with nocodazole, we investigated its effect on peak sodium current density of the human $\text{Na}_v1.2$ channel 5–24 h post-transduction (Figure 1B). Although the peak current amplitudes were increased in nocodazole-treated HEK293 cells (Figure S1B), there was no significant effect on their peak current density since the membrane capacitance of nocodazole-treated cells, which provides an accurate measure of cell surface area, was similarly increased (Figures S1C and S1D).

Effects of cell-cycle arrest at G1/S boundary on $\text{hNa}_v1.2$ expression

We then tested the effect of cell-cycle arrest at the G1/S boundary with thymidine (Figure 1A). Treating HEK293 cells with thymidine starting 24 h before the BacMam viral transduction of human $\text{Na}_v1.2$ resulted in 5- and 10-fold enhanced fluorescence signals of GFP compared with untreated transduced cells and lipofectamine-transfected cells, respectively (Figure 1C). The enhanced fluorescence signal in BacMam-transduced thymidine-treated HEK293 cells was accompanied by ~3.5- and 7-fold increases in current density compared with untreated transduced cells and lipofectamine-transfected cells, respectively, with no change in the membrane capacitance (Figures 1D, S2A, and S2B). To be sure that the increase in current density was the result of an increased number of expressed ion channels rather than an effect of thymidine on $\text{Na}_v1.2$ conductance, we measured gating currents that reflect the outward movements of the arginine gating charges of the channel protein and provided a proxy for the number of voltage-gated ion channels that are inserted in the plasma membrane. We found that

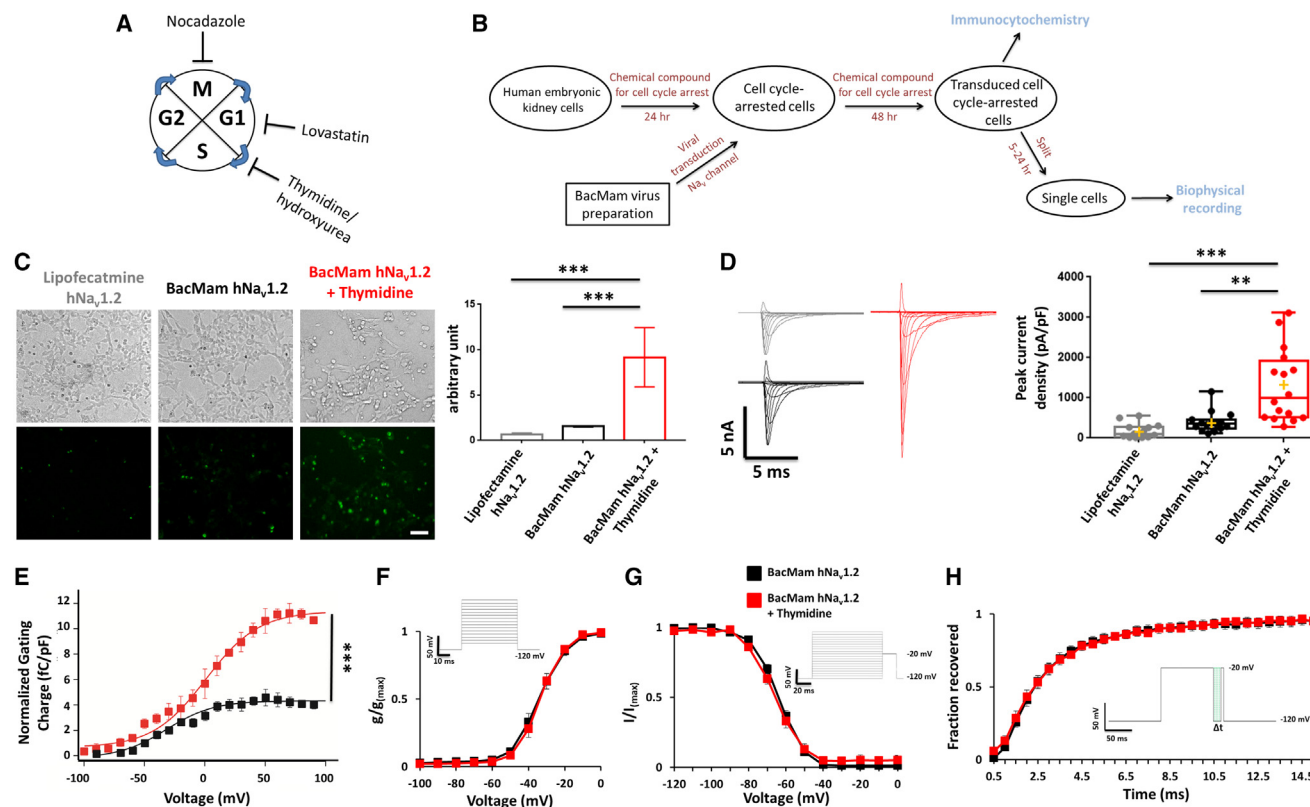


Figure 1. Enhanced expression of human $Na_V1.2$ in thymidine-treated HEK293 cells

(A) Schematic representation of the eukaryotic cell cycle highlighting the chemical compounds that can arrest the growth of cells at different stages. (B) A simple diagram of the workflow for expressing Na_V channels in mammalian cells. (C) Comparison of GFP fluorescence signal in HEK293 cells transfected with human $Na_V1.2$ cDNA vs. thymidine-untreated and thymidine-treated HEK293 cells after 48 h of viral transduction of the human $Na_V1.2$. Viral transduction in thymidine-treated cells resulted in approximately 5- and 10-fold enhanced fluorescence signals of GFP compared with untreated virally transduced and lipofectamine-transfected cells, respectively. The scale bar is 100 μm , and two-way ANOVA followed by Bonferroni's post hoc test was used. (D) Left: representative traces of peak current of human $Na_V1.2$ in lipofectamine-transfected cells and in thymidine-untreated and thymidine-treated virally transduced cells. Right: 3.5- and 7-fold increases in the peak current density of human $Na_V1.2$ in thymidine-treated HEK293 cells after 48 h of the viral transduction compared with untreated virally-transduced and lipofectamine-transfected cells, respectively. One-way ANOVA on ranks followed by Dunn's post hoc test was used. (E) Treating HEK293 cells with thymidine increased gating charge movement of the human $Na_V1.2$ after 48 h of viral transduction. $n = 9$ and 10 for $hNa_V1.2$ and $hNa_V1.2$ +thymidine, respectively. Unpaired, two-tailed Student's t test was used. (F) Similar steady-state activation curves of human $Na_V1.2$ in thymidine-treated and untreated HEK293 cells. (G) Similar steady-state inactivation curves of human $Na_V1.2$ in thymidine-treated and untreated HEK293 cells. (H) The recovery from fast inactivation of human $Na_V1.2$ was similar in thymidine-treated and untreated HEK293 cells. $**p < 0.01$, $***p < 0.001$. For (C) and (E)–(H), error bars indicate the standard error of the mean (SEM). For (D), boxes extend from the 25th to the 75th percentile of each group's distribution of values, with vertical lines and plus (+) signs denoting median and mean values, respectively. Vertical extending lines denote adjacent values (i.e., the most extreme values within the 1.5 interquartile range of the 25th and 75th percentiles of each group).

thymidine increased the gating charge movement measured from treated cells by more than ~ 3 -fold compared with the untreated controls (Figure 1E). To check if thymidine changes the biophysical profile of treated cells, we measured the steady-state activation and inactivation curves as well as the recovery from fast inactivation of sodium currents in thymidine-treated cells and compared them with untreated control cells. Notably, the activation, inactivation, and recovery from fast inactivation curves of both thymidine-treated and untreated HEK293 cells had similar $V_{1/2}$, k_v , and τ_{rec} values, confirming that thymidine does not modify the voltage-dependent gating of $Na_V1.2$ channels (Figures 1F–1H; Table 1).

Advantages of using thymidine over sodium butyrate to enhance Na_V expression

Sodium butyrate is widely used to increase the expression of different Na_V channels in cell culture preparations.^{24–27} To confirm whether cell-cycle arrest at the G1/S boundary can have an additive effect on increasing the expression of $Na_V1.2$, we compared the effect of sodium butyrate alone and in addition to thymidine. Treating cells with sodium butyrate alone and together with thymidine showed 3- and 6-fold increases in the fluorescence signal of GFP, respectively, compared with non-treated transduced cells (Figure S3A). Interestingly, although the expression of the human $Na_V1.2$

Table 1. Biophysical properties of Na_v channels in HEK293 and tsA201 cells

	Steady-state activation		Steady-state inactivation		Recovery from fast inactivation
	$V_{1/2}$ (mV)	K_v	$V_{1/2}$ (mV)	K_v	τ_{rec} at -120 mV (ms)
HEK293 cells					
BacMam h $\text{Na}_v1.2$	-34.38 ± 1.91	-6.70 ± 0.52	-64.02 ± 1.11	6.53 ± 0.12	2.08 ± 0.18
BacMam h $\text{Na}_v1.2$ + Thymidine	-33.68 ± 2.4	-6.19 ± 0.46	-63.44 ± 2.84	6.69 ± 0.21	2.29 ± 0.28
BacMam h $\text{Na}_v1.2$ + $\beta 1$	-27.74 ± 0.76	-5.12 ± 0.19	-61.55 ± 1.05	5.96 ± 0.21	1.78 ± 0.13
BacMam h $\text{Na}_v1.2$ + $\beta 1$ +thymidine	-27.27 ± 0.61	-5.31 ± 0.21	-63.27 ± 1.34	6.19 ± 0.31	2.01 ± 0.18
BacMam r $\text{Na}_v1.5$	-45.58 ± 2.96	-10.54 ± 1.34	N/A	N/A	N/A
BacMam r $\text{Na}_v1.5$ +thymidine	-44.83 ± 3.26	$-7.12 \pm 0.87^*$	N/A	N/A	N/A
tsA201 cells					
BacMam h $\text{Na}_v1.2$	-34.70 ± 2.49	-3.44 ± 0.54	N/A	N/A	N/A
BacMam h $\text{Na}_v1.2$ +thymidine	-37.39 ± 3.27	-3.69 ± 0.60	N/A	N/A	N/A
BacMam r $\text{Na}_v1.5$	-52.89 ± 1.91	-8.97 ± 0.46	N/A	N/A	N/A
BacMam r $\text{Na}_v1.5$ +thymidine	-55.76 ± 1.64	-8.95 ± 0.30	N/A	N/A	N/A

Data are presented as means $\pm$ SEM. Unpaired, two-tailed Student's *t* test was used. **p* < 0.05. $V_{1/2}$, the voltage of half-maximal activation; k_v , a slope factor; τ_{rec} , the time course of recovery from fast inactivation at -120 .

channel increased with the use of sodium butyrate, there was a corresponding decrease in its trafficking to the cell membrane, resulting in a tendency toward reduced peak current density, which was counteracted by the additive effect of thymidine (Figure S3B). The reduced peak current density of the h $\text{Na}_v1.2$ channel using sodium butyrate was also confirmed on another mammalian Na_v channel (rat $\text{Na}_v1.5$) (Figure S3C), highlighting the unique effect of cell-cycle arrest using thymidine on increasing both the expression of Na_v channels and their trafficking to the cell membrane.

Enhanced expression of $\text{Na}_v1.2$ channel with $\beta 1$ subunits by cell-cycle arrest

We tested the effect of thymidine on the expression of the human $\text{Na}_v1.2$ with the auxiliary $\beta 1$ subunit, a non-pore-forming subunit that is co-expressed with $\text{Na}_v1.2$ in nerve and muscle cells and affects expression, trafficking, and kinetics of human $\text{Na}_v1.2$.²⁸ The co-infection of both subunits in thymidine-treated HEK293 achieved a peak current density of 3,000 pA/pF (Figure 2A). The thymidine-treated cells showed a ~ 5 -fold increase in gating charge density compared with untreated cells, confirming the high expression of the functional human $\text{Na}_v1.2$ channel and its localization to the cell membrane (Figure 2B). This 5-fold increase in gating charge movement using thymidine with the combination of $\text{Na}_v1.2$ and $\beta 1$ subunits, compared with the previous ~ 3 -fold increase in the absence of the $\beta 1$ subunit (Figure 1E), may indicate the ability of thymidine to increase the expression and trafficking of the $\beta 1$ subunit as well. The increase in current and gating charge density by thymidine treatment was not accompanied by a change in the activation, inactivation, or recovery from fast inactivation profiles (Figures 2C–2E; Table 1). To evaluate the strength of our method to enhance the expression of a known loss-of-function mutation in the $\text{Na}_v1.2$ channel, which was previously shown to decrease the current density by 60%,²⁹ we transduced thymidine-treated HEK293 cells with $\text{Na}_v1.2$ (R853Q) alone and in combination with the auxiliary $\beta 1$ subunit. Compared to the current literature using the calcium phosphate transfection method, treat-

ing HEK293 cells with thymidine increased the current density by 5- and 10-fold using the BacMam viral transduction of $\text{Na}_v1.2$ (R853Q) and $\text{Na}_v1.2$ (R853Q)+ $\beta 1$ subunit, respectively (Figure 2F). These results indicate that the effects of cell-cycle arrest in increasing the expression of wild-type or mutant Na_v channels are not occluded by the presence of the $\beta 1$ subunit.

Enhanced expression of other mammalian Na_v channels in HEK293 cells with G1/S arrest

In order to determine whether the expression of other Na_v channels would be increased by thymidine cell-cycle arrest, we carried out similar experiments with the rat cardiac sodium channel, r $\text{Na}_v1.5$ (Figures 3A–3C). GFP fluorescence and peak sodium current were increased 2- to 2.5-fold, with no change in the voltage dependence of activation (Figures 3A–3C, S4A, and S4B; Table 1). These results indicate that expression of a different sodium channel from another mammalian species is similarly increased by cell-cycle arrest with thymidine. Hydroxyurea also arrests the cell cycle at the G1/S boundary (Figure 1A). To test whether cell-cycle arrest with hydroxyurea increased the human $\text{Na}_v1.5$ expression, we compared the expression of the h $\text{Na}_v1.5$ channel in hydroxyurea-treated and thymidine-treated HEK293 cells. Cell-cycle arrest with hydroxyurea or thymidine gave similar increases in GFP fluorescence and sodium current, with no change in cell capacitance, consistent with the conclusion that cell-cycle arrest at the G1/S boundary is the mechanism of action of both chemical compounds (Figures 3D, 3E, S4C, and S4D).

Enhanced expression of Na_v channels in tsA201 cells with G1/S arrest

We extended our analysis to a second HEK-derived cell line, tsA201, that is frequently used for the biophysical characterization of voltage-gated ion channels.³⁰ Similar to our results in HEK293 cells, the viral transduction of the human $\text{Na}_v1.2$ in thymidine-treated tsA201 cells enhanced GFP fluorescence by ~ 2.5 -fold (Figure 4A), which was accompanied by an increase in the peak

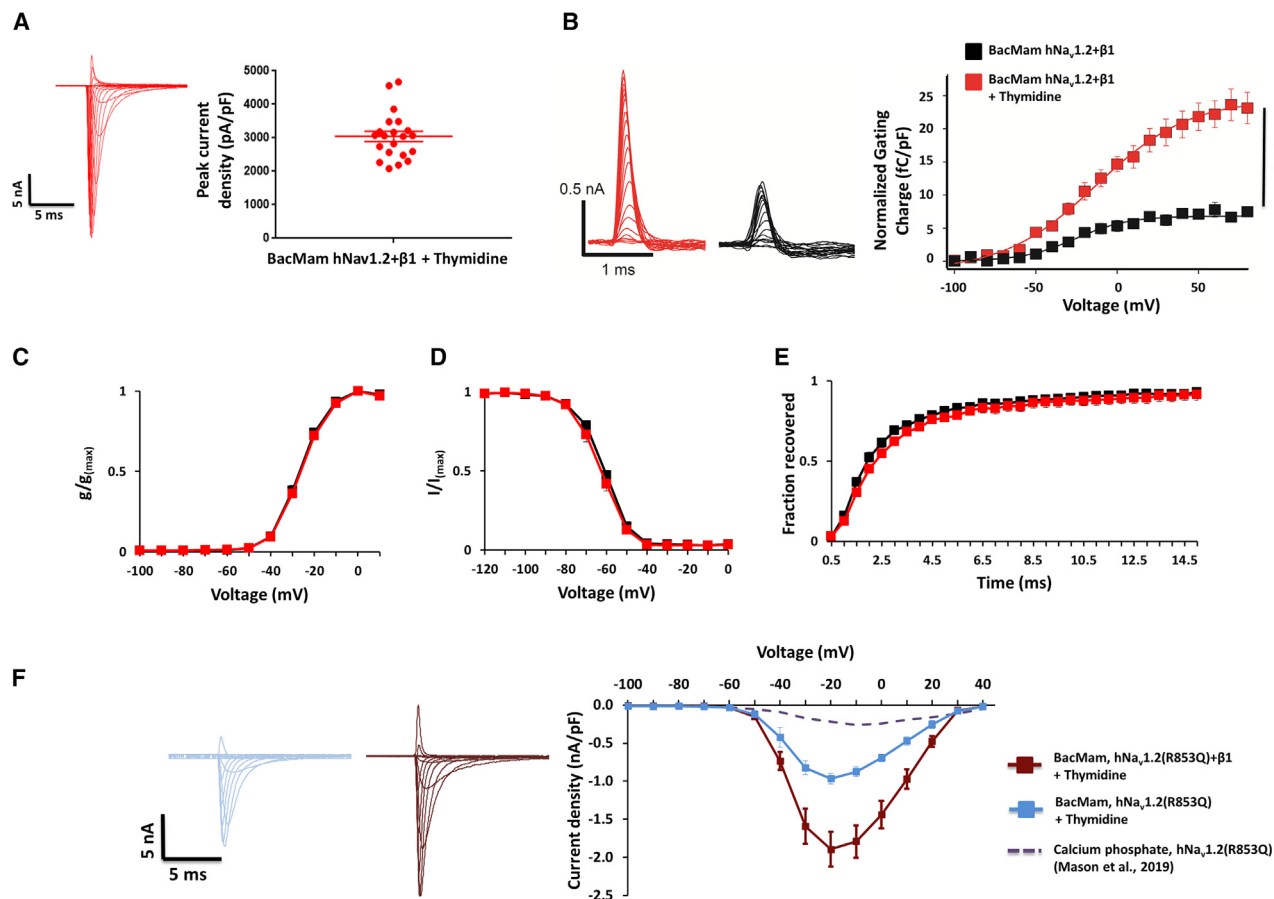


Figure 2. Enhanced expression of human $Na_v1.2$ with $\beta1$ subunit in thymidine-treated HEK293 cells

(A) Left: representative sodium currents conducted by $Na_v1.2$ channel expressed with $\beta1$ subunits with a ratio of 5:1 in thymidine-treated HEK293 cells. Right: the peak current density of the human $Na_v1.2$ in thymidine-treated HEK293 cells after a co-transduction with $\beta1$ subunit.

(B) Left: representative traces of gating currents of $hNa_v1.2+\beta1$ (black) and $hNa_v1.2+\beta1$ +thymidine (red). Right: treating HEK293 cells with thymidine increased gating currents of the human $Na_v1.2+\beta1$ after 48 h of viral transduction. $n = 5$ and 6 for $hNa_v1.2+\beta1$ and $hNa_v1.2+\beta1$ +thymidine, respectively. Unpaired, two-tailed Student's t test was used.

(C) Similar steady-state activation curves of human $Na_v1.2$ co-transduced with $\beta1$ subunit in thymidine-treated and untreated HEK293 cells.

(D) Similar steady-state inactivation curves of human $Na_v1.2$ co-transduced with $\beta1$ subunit in thymidine-treated and untreated HEK293 cells.

(E) The recovery from fast inactivation of human $Na_v1.2$ co-transduced with $\beta1$ subunit was similar in thymidine-treated and untreated HEK293 cells.

(F) Left: representative sodium currents conducted by the human $Na_v1.2(R853Q)$ channel expressed alone or with $\beta1$ subunit with a ratio of 5:1 in thymidine-treated HEK293 cells. Right: current-voltage curve of the human $Na_v1.2(R853Q)$ alone or co-transduced with $\beta1$ subunit in thymidine-treated HEK293 cells. The peak current density using this method was increased 5- to 10-fold compared to the literature using calcium phosphate transfection.²⁹ *** $p < 0.001$. Error bars indicate the SEM.

current amplitude and density, with no change in the membrane capacitance of cells (Figures 4B, S5A, and 5B). Similar to our results in HEK293 cells (Figure 1), we found no change in the steady-state activation parameters of human $Na_v1.2$ (Figure 4C; Table 1). For $rNa_v1.5$, the viral transduction of G1/S-arrested cells revealed a ~4-fold increase in the GFP fluorescence signal compared with non-arrested cells (Figure 4D). The enhanced fluorescence signal in G1/S-arrested cells was accompanied by an increase in the current amplitude and density, with no change in the membrane capacitance of cells (Figures 4E, S5C, and 5D). Similar to the human $Na_v1.2$, there was no difference in the steady-state activation parameters of the rat $Na_v1.5$ between thymidine-treated and untreated cells (Figure 4F; Table 1).

Enhanced expression of K_v channels in insect cells with G1/S arrest

To test whether G1/S arrest can enhance the transient expression of other ion channels in non-mammalian cell lines, we transduced Hi-5 insect cells with the human potassium $K_v7.3$ channel with and without thymidine using baculovirus containing the expression cassette of the pFastBac vector²¹ (Figure 5A). Treating Hi-5 cells with thymidine increased the current density of the $K_v7.3$ channel by ~2.5-fold at +100 mV without a change in cell membrane capacitance (Figures 5B and 5C). We then extended our analysis to a second insect cell line, Sf9, which showed a ~2-fold increase in the current density of the bacterial potassium K_v7Oc channel at +100 mV using thymidine compared with non-treated cells (Figure 5D).

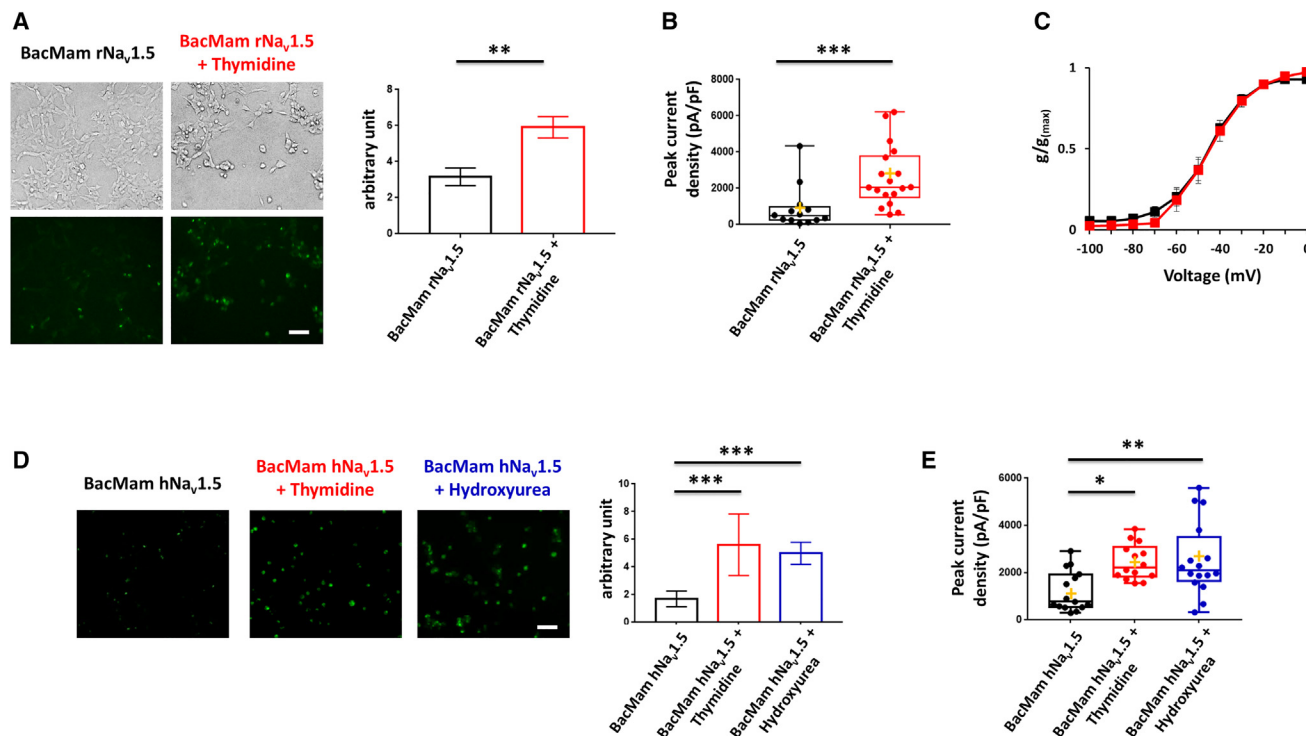


Figure 3. Enhanced expression of mammalian Na_v1.5 in HEK293 cells arrested at the G1/S boundary

(A) A 2-fold enhanced GFP fluorescence signal 48 h after the transduction of rat Na_v1.5 in thymidine-treated compared with untreated HEK293 cells.

(B) Cell-cycle arrest using thymidine increased the peak current density of rat Na_v1.5 by ~2.5-fold after 48 h of viral transduction. Unpaired, two-tailed Mann-Whitney test was used.

(C) Similar steady-state activation curves of rat Na_v1.5 in thymidine-treated and non-treated HEK293 cells.

(D) Comparison of GFP fluorescence signal in thymidine-treated, hydroxyurea-treated, and untreated HEK293 cells after 48 h of the viral transduction of human Na_v1.5. Both thymidine and hydroxyurea increased the GFP signal by 2.5-fold.

(E) Increased peak current density of human Na_v1.5 in thymidine-treated and hydroxyurea-treated cells compared with untreated HEK293 cells. One-way ANOVA followed by Tukey's post hoc test was used. **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

For (A) and (D), the scale bar is 100 μm, and two-way ANOVA followed by Bonferroni's post hoc test was used. For (A), (C), and (D), error bars indicate the SEM.

Similar to the aforementioned results in Hi-5 cells, there was no change in cell membrane capacitance using thymidine (Figure 5E).

The effect of thymidine cell-cycle arrest on soluble protein expression

The expression of membrane proteins can be more challenging than soluble proteins since they must be correctly folded and inserted into the lipid bilayer of the membrane. Still, the transient expression of soluble proteins in mammalian cell lines can encounter several hurdles that restrict the expression levels or functionality of the recombinant protein. Therefore, we investigated whether the aforementioned effect of cell-cycle arrest at the G1/S boundary in increasing the expression of ion channels, as an example of membrane proteins, can be extended to soluble proteins. We transduced HEK293 cells with the BacMam viral system expressing the histone demethylase enzyme attached to a reporter GFP protein. Treating the HEK293 cells with either thymidine or nocodazole showed a ~3.5-fold increase in the GFP fluorescence signal compared with non-treated cells, with no effect of lovastatin treatment (Figure 6). These results confirm that cell-cycle-arrest methods show the same effect on the expression of membrane and soluble proteins.

DISCUSSION

Cell-cycle arrest at G1/S boundary increases expression of Na_v and K_v channels

The significance of using cell protein expression systems in basic research and clinical applications is undisputed. However, the expression of complex membrane proteins, such as voltage-gated ion channels, often shows low protein yield and small functional signals owing to low copy numbers of protein expressed on the cell surface.¹² In this study, we have introduced cell-cycle arrest before and during BacMam viral transduction as a tool for increasing the expression of voltage-gated sodium channels in two HEK cell lines. Together, this combination of high-level viral expression and cell-cycle arrest at the G1/S boundary yields a ~7-fold increase in cell surface expression of Na_v1.2 and Na_v1.5 channels compared with transient expression of cDNAs in the same cell lines.^{31–34} Because Na_v1.2 and Na_v1.5 are from the tetrodotoxin-sensitive and -resistant branches of the Na_v channel family, respectively, these results indicate broad utility for these methods in expression of other Na_v channels. By extending our analysis to test K_v channels in non-mammalian cell lines, our findings demonstrated that

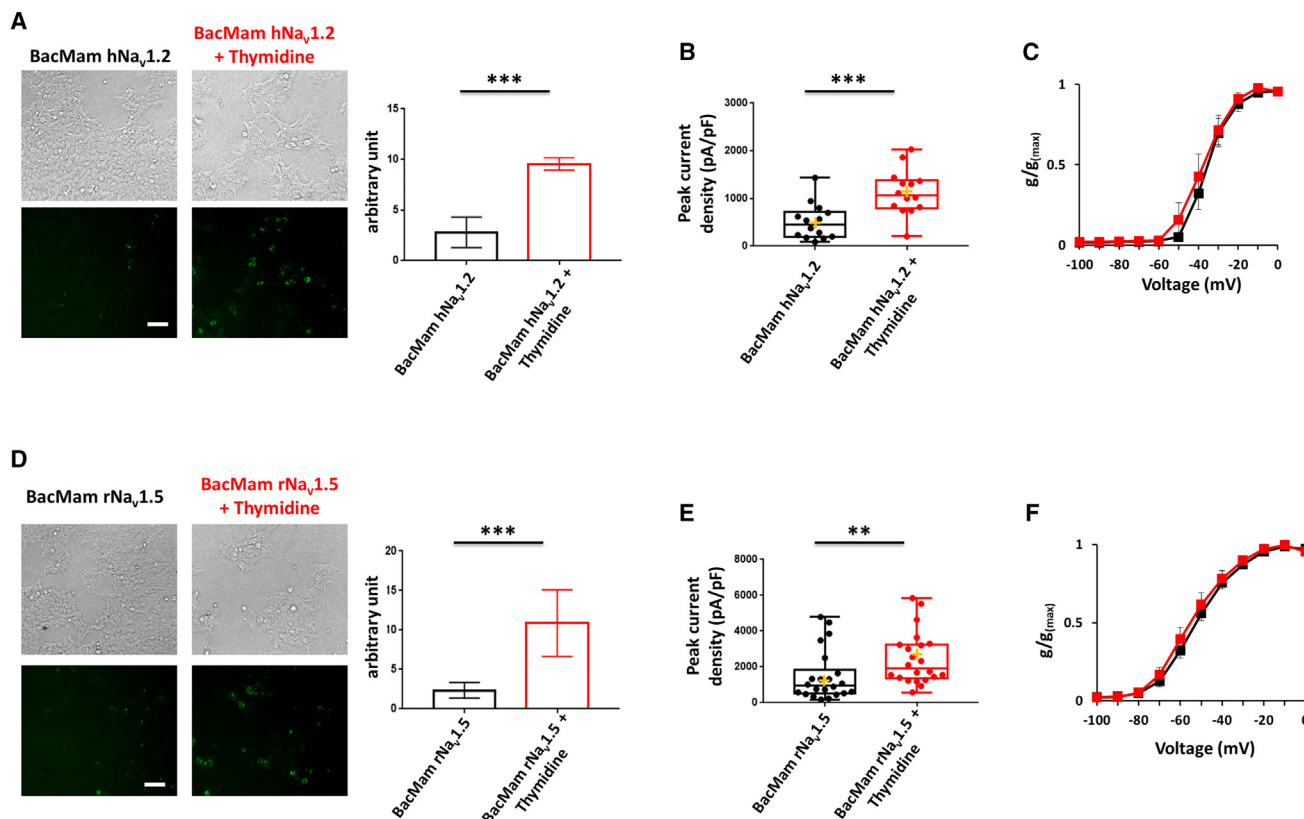


Figure 4. Enhanced expression of different mammalian Na_V channels in thymidine-treated tsA201 cells

(A) Comparison of GFP fluorescence signal in thymidine-treated vs. non-treated tsA201 cells after 48 h of viral transduction of human $\text{Na}_V1.2$. Thymidine increased the GFP expression by ~ 3 -fold.
 (B) A 2-fold increase in the peak current density of the human $\text{Na}_V1.2$ channel in thymidine-treated tsA201 cells compared with non-treated cells. Unpaired, two-tailed Student's t test was used.
 (C) Similar steady-state activation curves of the human $\text{Na}_V1.2$ channel in thymidine-treated and untreated tsA201 cells.
 (D) Comparison of GFP fluorescence signal in thymidine-treated and non-treated tsA201 cells after 48 h of transduction of the rat $\text{Na}_V1.5$. Thymidine increased the GFP expression by ~ 3.5 -fold.
 (E) A 2-fold increase in the peak current density of the rat $\text{Na}_V1.5$ in thymidine-treated tsA201 cells after 48 h of viral transduction. Unpaired, two-tailed Mann-Whitney test was used.
 (F) Similar steady-state activation curves of the rat $\text{Na}_V1.5$ in thymidine-treated and non-treated tsA201 cells. ** $p < 0.01$, *** $p < 0.001$.
 For (A) and (D), the scale bar is $100 \mu\text{m}$, and two-way ANOVA followed by Bonferroni's post hoc test was used. For (A), (C), (D), and (F), error bars indicate the SEM.

inducing cell-cycle arrest at the G1/S boundary can increase the expression of human and bacterial K_V channels in two insect cell lines, Hi-5 and Sf9. These results suggest that our method may be widely applicable and may serve as a universal protocol to improve the transient expression of voltage-gated ion channels, and possibly other proteins, in various mammalian and non-mammalian cell lines. Additional research on expression of other membrane and soluble proteins in other cell lines is required to validate this proposal.

Cell-cycle arrest is a well-established method for cell-cycle synchronization to analyze various aspects of cellular metabolism at specific stages of the cell cycle and for the study of DNA damage repair mechanisms.^{35–37} Cell-cycle arrest can be performed at different stages of the cell cycle (G1, S, and M stages) (for a review, see Ligasová and Koberna³⁶), but we found that arrest at the G1/S boundary was uniquely effective.

Cell-cycle arrest at other boundaries is not effective

To test cell-cycle arrest at other boundaries, we first tried arresting HEK293 cells at the M stage using nocodazole, which is known to inhibit mitotic spindle formation.³⁸ We found an increase in GFP fluorescence and sodium current amplitude of h $\text{Na}_V1.2$; however, the enhancement in protein expression and function was accompanied by a comparable increase in cell membrane capacitance and surface area owing to the inhibition of cell division. Therefore, this method does not increase the density of functional Na_V channels on the cell surface.

Cell-cycle arrest of HEK293 cells in the G1 phase using lovastatin also did not reveal an enhanced effect on the expression of the GFP reporter protein. The mechanism of lovastatin arrest at the G1 phase is not completely understood,³⁹ but several studies have shown that lovastatin can cause apoptosis of

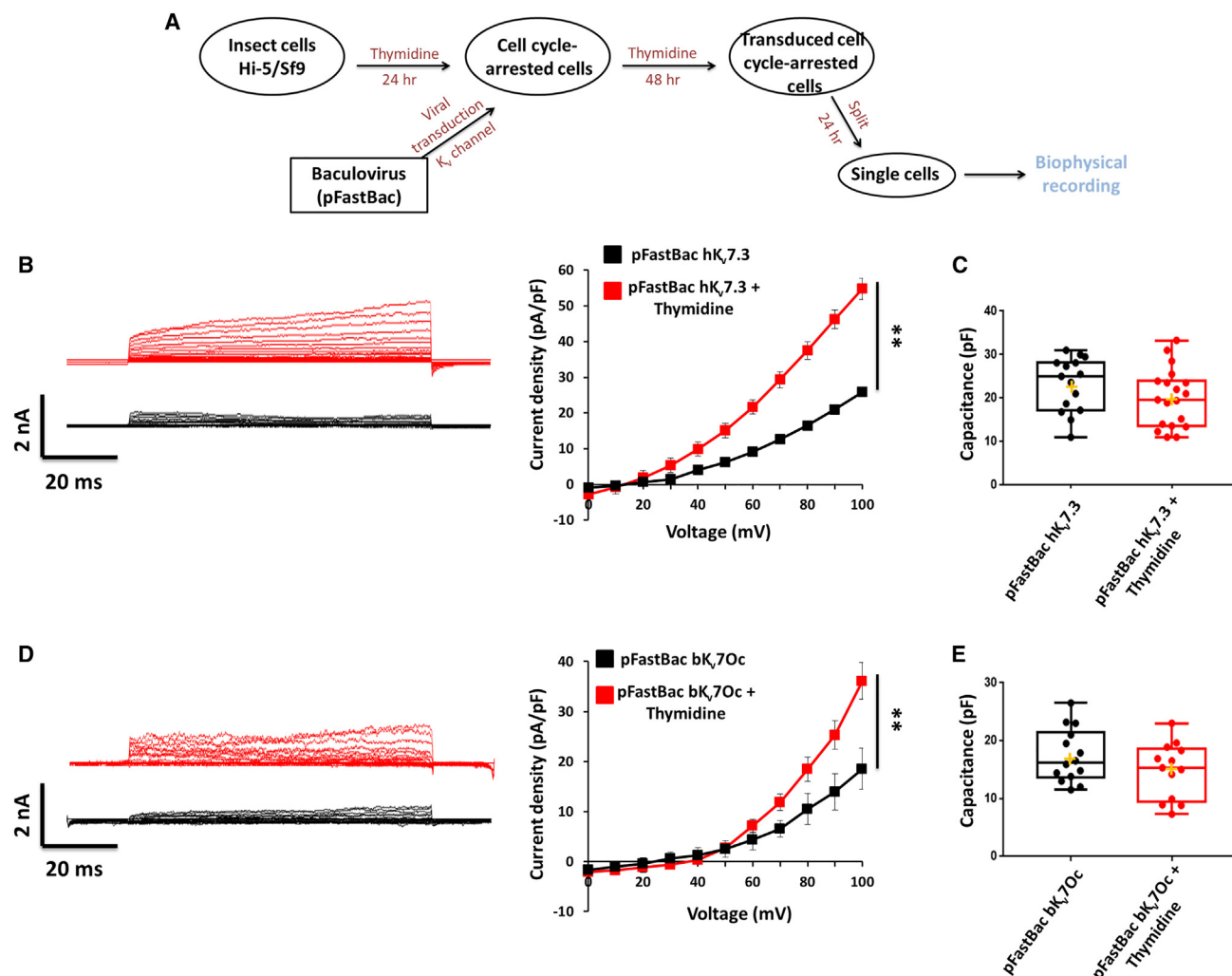


Figure 5. Enhanced expression of human and bacterial potassium Kv7 channels in thymidine-treated Hi-5 and Sf9 insect cells

(A) A simple diagram of the workflow for expressing Kv7 channels in insect cells.

(B) Left: representative potassium currents conducted by human Kv7.3 channel in thymidine-treated and untreated Hi-5 cells. Right: current-voltage curve of the human Kv7.3 channel in thymidine-treated and untreated Hi-5 cells. Thymidine increased the current density of the Kv7.3 channel by 2.5-fold at +100 mV.

(C) No difference in the capacitance of thymidine-treated and non-treated Hi-5 cells after 48 h of the human Kv7.3 viral transduction.

(D) Left: representative potassium currents conducted by bacterial Kv7Oc channel in thymidine-treated and untreated Sf9 cells. Right: current-voltage curves of the human Kv7.3 channel in thymidine-treated and untreated Sf9 cells. Thymidine increased the current density of the Kv7Oc channel by 2-fold at +100 mV.

(E) No difference in the capacitance of thymidine-treated and non-treated Sf9 cells after 48 h of the bacterial Kv7Oc viral transduction.

For (B) and (D), $p < 0.01$, and error bars indicate the SEM. For (C) and (E), boxes extend from the 25th to the 75th percentile of each group's distribution of values with vertical lines and + signs denoting median and mean values, respectively. Vertical extending lines denote adjacent values (i.e., the most extreme values within the 1.5 interquartile range of the 25th and 75th percentile of each group).

multiple cell types.^{40–43} This may explain why there was a trend toward a decreased overall expression of the GFP reporter protein in our lovastatin-treated HEK293 cells.

Mechanisms of cell-cycle arrest at the G1/S boundary

Cell-cycle arrest at the G1/S boundary can be achieved using different compounds, including thymidine, hydroxyurea, aphidicolin, and methotrexate, which frequently target the synthesis of deoxyribonucleotides.³⁶ Thymidine blocks DNA replication by interrupting the deoxynucleotide metabolism pathway through competitive inhibition.³⁶ On the other hand, hydroxy-

urea halts DNA synthesis by inhibiting the ribonucleotide reductase enzyme, thus reducing the production of dNTPs.⁴⁴ The reduced concentration of dNTPs affects the function of DNA polymerase at replication forks.⁴⁵ Treating cells with thymidine or hydroxyurea 24-h before the viral transduction of voltage-gated ion channels increased their expression, possibly by stopping DNA replication and redirecting cell metabolism into RNA and protein expression and thereby enhancing cellular production of recombinant proteins. Since DNA synthesis inhibition causes a reduction in the growth rate of cells, our results are consistent with the enhancement of expression of

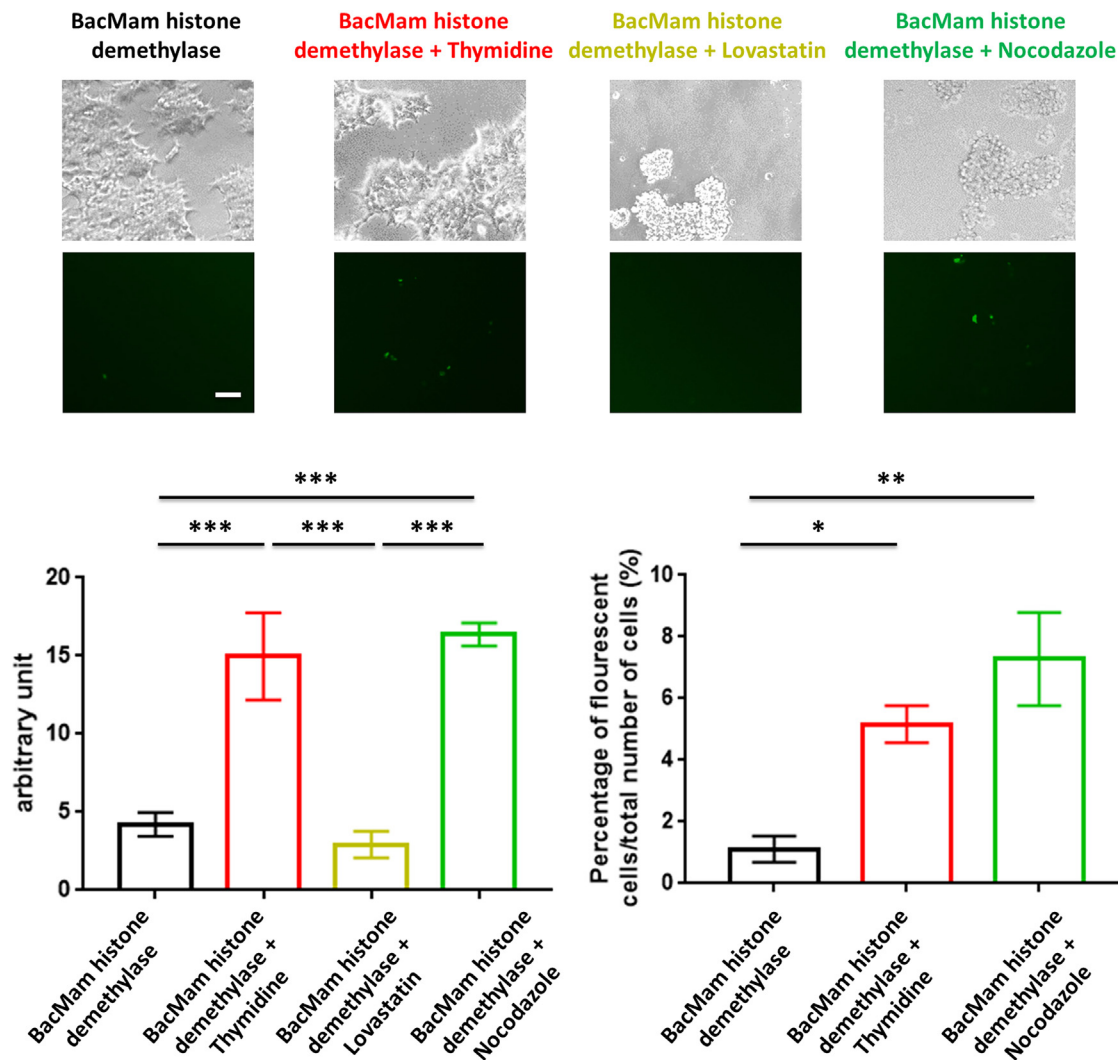


Figure 6. Enhanced expression of histone demethylase enzyme in thymidine- and nocodazole-treated HEK293 cells

Comparison of GFP fluorescence signal in thymidine-, lovastatin-, and nocodazole-treated HEK293 cells after 48 h of the viral transduction of the histone demethylase enzyme attached to GFP as a marker for expression. Both thymidine and nocodazole significantly increased the GFP expression by ~3-fold and increased the percentage of fluorescent cells.

heterologously expressed proteins by decreasing the cell culture temperature, which has previously been shown to reduce the growth rate of cells.¹² Compared to the current literature, the combination of cell-cycle arrest at the G1/S boundary and the BacMam viral system revealed a ~7-fold increase in the current density of different voltage-gated sodium channels in both HEK293 and tsA201 cells,^{31–34} confirming the strength and the wide application of our protocol. By co-transducing both human Na_v1.2 and the β 1 subunit with a ratio of 5:1, we were able to increase the peak current density of the human Na_v1.2 to 3 nA/pF, which represents a 5-fold increase compared to a previous study in HEK293 cells using lipofectamine transfection.⁴⁶ The increase in the gating charges in G1/S-arrested cells confirms that the mechanism of action of thymidine/hydroxyurea is enhancing the expression of sodium

channels without changing single-channel conductance. This shows the suitability of our modified method of viral transduction/cell-cycle arrest to measure tiny gating currents or gating pore currents that are in the range of 0.1%–1% of the peak sodium current of voltage-gated ion channels. Measuring such important biophysical features is mandatory to fully characterize the effect of some mutations and their pathophysiological mechanisms related to human disorders. Interestingly, the similarity of steady-state activation parameters of different mammalian sodium channels in G1/S-arrested and unarrested cells indicates that the functionality of the expressed sodium channels was not affected by treatment with thymidine.

The possible effect of thymidine in stopping DNA replication and redirecting cell metabolism into RNA and protein expression was also apparent in the non-mammalian cell lines Hi-5 and Sf9.

We expect that the effect of thymidine on increasing the transient expression level of proteins depends on the duration of the cell cycle and on the doubling times of cells as well as on the size and complexity of the expressed protein. For example, Sf9 doubling times vary between 24 and 30 h, while Hi-5 is between 18 and 24 h.⁴⁷ This indicates that Sf9 cells may require a longer duration of treatment with thymidine than Hi-5 cells to obtain a similar level of enhancing the expression of the same protein. Further research is necessary to confirm this hypothesis. Worth noticing is the difference in the expression levels of proteins according to the used cell lines. For example, higher protein yields are known to be achieved in Hi-5 compared with Sf9 cells.^{48,49} Hence, it is important to consider these factors while selecting a suitable cell line to achieve the maximum expression of recombinant proteins.

Loss-of-function mutations

Our approach will help to resolve a long-lasting problem in the field of voltage-gated ion channels concerning the subset of mutations tagged with the name “loss of function.” These mutations are defined by this name because of their very low current density. Some of these mutations affect the folding or the trafficking of the channels. However, others have similar surface expression compared with the wild-type channel but show lower current density. The very low expression of those channels hinders the ability to resolve their biophysical characteristics. To test that, we expressed a mutation, R853Q, that is known to lower the surface expression of Na_v1.2 by 60%. Using our expression method, we were able to retrieve the full expression of the channel and thus study the functional properties of the mutant.

Comparison with other expression systems

Other efforts to increase the expression levels of proteins in different heterologous mammalian models have shown some success but with limits. Optimization of the vector system using a strong promoter, a proper signal peptide, and codon tailoring has shown increased protein expression from poorly translated transcripts.^{10,50} One study revealed that the multiplicity of viral transduction of HeLa cells allowed recombinant gene expression to be prolonged and increased protein expression in the short term.⁵¹ In another study, the addition of sodium butyrate as a non-specific inhibitor of histone deacetylase has shown an increase in the virus-mediated gene expression in several cell types.^{52,53} However, in our hands, the increase in the expression of Na_v channels by 3-fold using sodium butyrate negatively affected their trafficking to the cell membrane and, hence, reduced the recorded current density, which was counteracted by the additive effect of thymidine (Figure S3).

The culture temperature of mammalian cell lines has also been demonstrated to play a role in protein expression. Incubation of cells at a lower temperature of 34°C rather than 37°C was used to increase protein yields in transduced Chinese hamster ovary (CHO) cells grown in suspension culture.⁵³ Similarly, reducing the culture temperature of HEK293S cells to 33°C caused a 1.5-fold higher expression of GFP and α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptors.¹²

In another study, a 5 min exposure of 10% DMSO to HEK293 cells revealed a $\sim$ 1.6-fold increase in the GFP fluorescence signals without causing any appreciable cytotoxicity.⁵⁴ Nevertheless, the methods we describe here give a much greater increase in the expression of the reporter GFP and voltage-gated ion channels in cells along with their trafficking to the cell membrane and may therefore be ideal for their biophysical characterization.

In our study, we showed that using thymidine does not affect the biophysical properties of the expressed proteins, as indicated by the similarity of steady-state activation and inactivation curves, as well as the recovery from fast inactivation in treated and untreated cells. However, using chemical compounds to arrest cells at specific stages is known to result in unwanted effects on cellular metabolism.³⁶ Trapping cells in the G1/S boundary causes exposure to replication stress as their replication forks are stalled, which can cause double-stranded DNA breaks.⁵⁵ Additionally, chromosomal aberrations were previously observed after the use of thymidine treatment.⁵⁶ In our hands, hydroxyurea had a drastic effect on the health of HEK293 cells, as indicated by the number of dead cells floating in the medium. Nevertheless, we were able to grow enough cells per dish for fluorescence analysis and biophysical characterization of sodium channels. Still, the potential negative effects of these compounds must be taken into account when planning experiments. On the other hand, one advantage of the method we have described here is that it is simple to implement. We recommend preparing a fresh solution of thymidine or hydroxyurea for each viral transduction.

Limitations of the study

The methods described here give a substantial increase in the expression level of sodium and potassium channels as measured by the GFP marker and by electrophysiological recordings. Still, it is likely that the level of expression will be limited by the size and complexity of the proteins under study and that other cell types with different cell-cycle durations and membrane lipid components will not be as effective in achieving the maximum expression of ion channels, even after treatment with thymidine or hydroxyurea. Since HEK293 cells are also suitable for suspension culture in shaker flasks for large-scale production of proteins per unit culture volume, it will be of interest to test the effect of arresting the cell cycle at the G1/S boundary on the expression of recombinant proteins in suspension cells, which would enable more efficient analysis of the structure of ion channels with high resolution. However, since the cell-cycle arrest stops the growth rate of cells, we do not expect an increase in the total yield of protein per dish/flask compared with untreated cells, which could be a limitation for the structural investigation of voltage-gated ion channels.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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AUTHOR CONTRIBUTIONS

Conceptualization, A.E. and T.M.G.; data curation, A.E. and T.M.G.; formal analysis, A.E., W.A.C., and T.M.G.; validation, A.E., W.A.C., and T.M.G.; visualization, A.E.; writing – original draft, A.E.; writing – review & editing, A.E., W.A.C., and T.M.G.; funding, A.E. and W.A.C.; supervision, W.A.C. and T.M.G.; project administration, W.A.C. and T.M.G.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Cell lines		
HEK293	ATCC	CRL-1573™
tsA201	Millipore Sigma	96121229
<i>Spodoptera frugiperda</i> (Sf9)	Gibco	11496015
<i>Trichoplusia ni</i> (Hi-5)	Invitrogen	B85502
Bacterial and viral strains		
DH10bac cells	Gibco	10361012
Chemicals, peptides, and recombinant proteins		
Dulbecco's Modified Eagle's Medium (DMEM)	Corning	10-013-CV
DPBS	Gibco	14190-136
Trypsin-EDTA 0.25	Gibco	25200-056
Lipofectamine 3000	Invitrogen	L3000-015
Cellfectin-II	Gibco	10362-100
SOC medium	Quality Biological	340-031-671EA
Grace's insect medium	Gibco	11605-094
Fetal bovine serum	VWR	89510-186
L-Glut:Pen:Strep	Gemini Bio-product	400-110
Nocodazole	Sigma-Aldrich	M1404-2MG
Lovastatin/Mevinolin	Sigma-Aldrich	M2147-25MG
Thymidine	Sigma-Aldrich	T1895-25G
Hydroxyurea	Sigma-Aldrich	H8627-1G
Tetrodotoxin (TTX)	Tocris Bioscience	1078
NaCl	Sigma-Aldrich	CAS 7647-14-5
CsF	Sigma-Aldrich	289345-25G
EGTA	Sigma-Aldrich	CAS 67-42-5
HEPES	Sigma-Aldrich	CAS 7365-45-9
CaCl ₂	Sigma-Aldrich	CAS 10035-04-8
MgCl ₂	Macron Fine Chemicals	5958-04
KF	Sigma-Aldrich	402931-100G
N-Methyl-D-glucamine (NMDG)	Sigma-Aldrich	M2004-500G
Critical commercial assays		
E.Z.N.A.® Plasmid DNA Midi Kit	Omega, Bio-tek	D6904-03
Software and algorithm		
ImageJ 1.52a	NIH	https://imagej.nih.gov/ij/download.html
Clampfit software of pCLAMP 11.0.3	Axon Instruments	https://www.moleculardevices.com/products/axon-patch-clamp-system/acquisition-and-analysis-software/pclamp-software-suite
Igor Pro 6.37	WaveMetrics	https://www.wavemetrics.com/software/igor-pro-637-installer
Microsoft Excel	Microsoft Corporation	https://www.microsoft.com/en-us/microsoft-365/excel
GraphPad Prism 9	GraphPad Software	https://www.graphpad.com/scientific-software/prism/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Tamer M. Gamal El-Din, tmgamal@uw.edu.

Materials availability

All unique reagents used in this study, including BacMam and pFastBac vectors, are available from the [lead contact](#) with a completed Materials Transfer Agreement.

Data and code availability

- All data reported in this paper will be shared by the [lead contact](#) upon request.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell lines

HEK293 and tsA201 cells

HEK293 (ATCC, CRL-1573) and tsA201 (derived from the same line of HEK293 cells and stably transfected with the temperature-sensitive gene for SV40 T-antigen, Millipore Sigma, 96121229) cells were maintained at 37°C under 5% CO₂ in Dulbecco's Modified Eagle's Medium (DMEM) (Corning) supplemented with 10% FBS (VWR) and 1% L-Glutamine:Penicillin:Streptomycin solution (Gemini Bio-product).

Hi-5 and Sf9 cells

Trichoplusia ni (Hi-5) (Invitrogen, B85502) and *Spodoptera frugiperda* (Sf9) (Gibco, 11496015) cells were grown in Grace's insect medium (Gibco) supplemented with 10x fetal bovine serum (FBS) (VWR) and 1% L-Glutamine:Penicillin:Streptomycin solution (Gemini Bio-product).

METHOD DETAILS

Bacmid preparation

The human Na_v1.2, human Na_v1.5, rat Na_v1.5 and human β1 subunit clones were used for subcloning into the pEG-BacMAM¹⁸ containing the full-length human Na_v1.2, human Na_v1.5, rat Na_v1.5 and human β1 subunit, respectively, each attached to GFP as a marker for expression. The human K_v7.3 and bacterial K_v7Oc from *O. cyanobacterium* were subcloned into the pFastBac vector (Invitrogen) as previously described.²¹ The bacterial transformation of DNA was prepared using the standard procedure. Briefly, the PCR products were transformed into 50 μL DH10bac cells (Gibco, 11496015) to generate bacmid for protein expression and were incubated for 30 min on ice before cells were heat-shocked for 30 s at 42°C. Bacterial cells were recovered on ice for 2 min. After recovery, 250 μL of pre-warmed SOC medium (Quality Biological) were added for outgrowth, and cells were incubated at 37°C with shaking at 300 rpm for 4 h. Then, cells were plated onto agar plates containing tetracycline, kanamycin and gentamycin. Antibiotic-resistant colonies that emerged on the LB plates were selectively picked and inoculated in 8-mL cell culture tubes containing LB medium (10 g/L B-tryptone, 5 g/L Bactoyeast, 10 g/L NaCl with pH 7.0 and autoclaved) supplemented with tetracycline, kanamycin and gentamycin, and cultured overnight at 37°C. On the second day, midi-prep was performed to obtain plasmid DNA from the bacterial cultures using E.Z.N.A. Plasmid DNA Midi Kit (Omega, Bio-tek, D6904-03) according to the manufacturer's procedure.

Baculovirus generation

The recombinant baculoviruses were produced in insect cells from *Spodoptera frugiperda* (Sf9) as previously described.^{57,24} In brief, Sf9 insect cells were transfected with the bacmid of either human Na_v1.2, human Na_v1.5, rat Na_v1.5 or the human β1 subunit by Cellfectin-II reagent (Gibco), and the transfected cells were cultured in a 6-well dish for 72 h in a 27°C incubator to make P1 viruses. P1 viruses were then collected and transduced another batch of Sf9 cells in a 150 mm dish with 20 mL Grace's insect medium (Gibco) supplemented with 10x fetal bovine serum (FBS) (VWR) and 1% L-Glutamine:Penicillin:Streptomycin solution (Gemini Bio-product), and the transduced cells were cultured for 72 h in a 27°C incubator to make P2 viruses. After collecting the P2 viruses, they were used to transduce a new batch of Sf9 cells in a 150 mm dish, and the transduced cells were cultured for 72 h in a 27°C incubator to make P3 viruses that were used to express sodium channels in the mammalian cell line HEK293 and tsA20.

Transfection of Na_v1.2 channel in HEK293 cells

HEK293 cells were plated in 35-mm cell culture dishes with 70% confluency and then transiently transfected after 24 h with 1 μg Na_v1.2 cDNA using Lipofectamine 3000 Reagent (Invitrogen) according to the manufacturer's procedure. After transfections, cells

were incubated at 37°C in 5% CO₂ for 48 h before being investigated under the fluorescence microscope to check the intensity of the GFP fluorescence signal. Cells were then detached using trypsin-EDTA (Gibco), split into single cells and allowed to settle down and attach to the surface of the dish for 5–24 h before performing electrophysiological recording.

Transduction and expression of Na_v channels in HEK293 and tsA201 cells

Based on the increased protein expression in cells with a reduced growth rate, we tested the possibility of enhancing the expression of voltage-gated sodium channels by completely inhibiting the cell growth of human embryonic kidney cells, HEK293 and tsA201. Cells maintained at 37°C under 5% CO₂ were plated in 35-mm cell culture dishes to achieve 70% confluency 24 h before the transduction with the recombinant baculovirus. Either nocodazole (Sigma-Aldrich, M1404-2MG) (in DMSO), lovastatin/Mevinolin (Sigma-Aldrich, M2147-25MG) (in ethanol), thymidine (Sigma-Aldrich, T1895-25G) (in DPBS) or hydroxyurea (Sigma-Aldrich, H8627-1G) (in H₂O) was added to the culture directly after splitting the cells in 35 mm cell culture dishes with a final concentration of 100 ng/mL, 10 mM, 2 mM and 2 mM, respectively. After 24 h, the medium was removed from the dish and 0.5 mL of the virus (human Nav1.2, human Nav1.5, rat Nav1.5 or human Nav1.2+β1 with a ratio of 5:1) was added to the HEK293/tsA201 cell dish in addition to 0.5 mL fresh DMEM medium supplemented with 10x FBS and L-Glutamine:Penicillin:Streptomycin solution and were left for 1.5 h on a shaker. In another set of experiments, we added only 0.1 mL of the virus to 0.5 mL of the medium to get current amplitudes of less than 6 nA for running activation, inactivation and recovery from fast inactivation protocols. Another 1 mL of the same medium was added to the dish with nocodazole, lovastatin, thymidine or hydroxyurea, with a final concentration of 100 ng/mL, 10 mM, 2 mM and 2 mM, respectively. Cells were then cultured in a 37°C incubator. For testing the effect of sodium butyrate, 5 mM sodium butyrate were added to the culture after 6 h. Cells were cultured for 48 h before being investigated under the fluorescence microscope to check the intensity of the GFP fluorescence signal. Cells were then split into single cells and allowed to settle down and attach to the surface of the dish for 5–24 h before performing electrophysiological recording. A simple diagram of the workflow can be found in [Figure 1B](#).

Transduction and expression of K_v channels in Hi-5 and Sf9 cells

Hi-5 and Sf9 cells were grown on 35-mm Petri dishes in Grace's insect medium (Gibco) supplemented with 10x fetal bovine serum (FBS) (VWR) and 1% L-Glutamine:Penicillin:Streptomycin solution (Gemini Bio-product). Thymidine (Sigma-Aldrich, T1895-25G) (in DPBS) was added to the culture medium with a final concentration of 2 mM. After 24 h, cells were transduced by replacing the incubation medium with 0.5 mL of either human K_v7.3 or bacterial K_v7Oc viruses. After 1 h, 1.5 mL of incubation medium was added to the virus-containing medium. Cells were then maintained at 27°C for 48 h and split into single cells. They were allowed to settle down and attach to the surface of the dish for 24 h before performing electrophysiological recording. A simple diagram of the workflow can be found in [Figure 5A](#).

Electrophysiological recording

Sodium and potassium currents were recorded from transduced cells using the whole-cell configuration of the patch-clamp technique. Data acquisition was conducted using an Axopatch 200B amplifier (Molecular Devices), and voltage commands were generated using Pulse 8.5 software (HEKA, Germany), and ITC18 analog-to-digital interface (Instrutech, Port Washington, NY). Currents resulting from applied pulses were filtered at 5 kHz with a low-pass Bessel filter and then digitized at 20 kHz. Leak and capacitance transient currents were subtracted using a P/4 protocol. All recordings were performed at room temperature of 21°C–24°C. Data were analyzed using Clampfit software of pCLAMP 11.0.3 (Axon Instruments), Igor Pro 6.37 (WaveMetrics) and Microsoft Excel (Microsoft Corporation, Redmond, WA, USA).

Recording pipettes for measuring sodium currents were pulled from borosilicate glass to achieve initial bath resistances of 1.5–3.0 MΩ and filled with an intracellular solution containing (in mM): 35 mM NaCl, 105 mM CsF, 10 mM EGTA and 10 mM HEPES. The extracellular patch-clamp solution contained (in mM): 140 mM NaCl, 2 mM CaCl₂, 2 mM MgCl₂ and 10 mM HEPES. The pH of both intracellular and extracellular solutions was adjusted to 7.4 using a CsOH solution. Untreated HEK293 and tsA201 cell lines had an average 10 pF cell membrane capacitance (C_m) and series resistances (R_s) varied from 2 to 6 MΩ.

Recordings in transduced HEK293 and tsA201 cells were performed by holding the cells at –120 mV for 10 min after establishing the whole-cell configuration. Only cells with a maximal voltage error due to residual series resistance of less than 5 mV after 90% compensation were chosen for further evaluation. For peak current measurement, current-voltage (I/V) relationships were recorded in response to voltage steps (50 ms) ranging from –100 mV to +50 mV in 10 mV increments.

The activation curve (conductance versus voltage, G/V), inactivation and recovery from fast inactivation were obtained from cells showing a peak current amplitude of less than 6 nA to assure the fidelity of the voltage clamp. The steady-state activation curve was fitted by a Boltzmann function: $G/G_{\max}(V) = 1/(1 + \exp[(V - V_{1/2})/k_v])$, where the conductance (G) is calculated from $G = I/(V - V_{\text{rev}})$, $G_{\max}$: the maximal conductance, V_{rev} : the reversal potential of Na⁺, $V_{1/2}$: the voltage of half-maximal activation, and k_v is a slope factor.

Steady-state inactivation was determined using 100-ms conditioning pulses to various potentials ranging from –120 to +70 mV in 10 mV increments. In succession, the remaining sodium current was elicited via a 20 ms long test pulse at –20 mV. A standard Boltzmann function was fit to the inactivation curves: $I(V) = I_{\max}/(1 + \exp[(V - V_{1/2})/k_v])$, with I being the recorded current amplitude at the conditioning potential V, $I_{\max}$ being the maximal current amplitude, $V_{1/2}$ the voltage of half-maximal inactivation and k_v a slope factor.

Recovery from fast inactivation was recorded from holding potentials of –120 mV. Cells were depolarized to –20 mV for 100 ms to inactivate all voltage-gated Na⁺ channels and then repolarized to –120 mV for increasing duration followed by a second depolarizing

pulse to -20 mV for 5 ms. A first-order exponential function with an initial delay was fit to the time course of recovery from inactivation yielding the time constant τ_{rec} .

For gating current measurement, tetrodotoxin (TTX) was added to the extracellular solution with a final concentration of $1 \mu\text{M}$. Transduced HEK293 cells were held at -140 mV for 10 min after establishing the whole-cell configuration. Current-voltage (I/V) relationships were recorded in response to voltage steps (lasting for 20 ms) ranging from -100 mV to $+100$ mV in 10 mV increments. For quantitative comparison, we always integrated the gating current in response to a depolarizing step, ON gating current. The time integration was performed after the baseline at the end of the depolarization was subtracted from the trace.

For measuring potassium currents, recording pipettes were pulled from borosilicate glass to achieve initial bath resistances of $1.5\text{--}3.0 \text{ M}\Omega$ and filled with an intracellular solution containing (in mM): 140 mM KF, 10 mM EGTA and 10 mM HEPES. The extracellular patch-clamp solution contained (in mM): 140 mM NMDG-methanesulfonate, 2 mM CaCl_2 , 2 mM MgCl_2 and 10 mM HEPES. The pH of both intracellular and extracellular solutions was adjusted to 7.4 using a CsOH solution. Untreated Hi-5 and Sf9 cell lines had an average of 20 and 15 pF cell membrane capacitances (C_m), respectively and series resistances (R_s) varied from 2 to $6 \text{ M}\Omega$.

Recordings in transduced Hi-5 and Sf9 cells were performed by holding the cells at -100 mV for 1 min after establishing the whole-cell configuration. Only cells with a maximal voltage error due to residual series resistance of less than 5 mV after 90% compensation were chosen for further evaluation. Current-voltage (I/V) relationships were recorded in response to voltage steps (100 ms) ranging from -70 mV to $+100$ mV in 10 mV increments. Data are shown from 0 to $+100$ mV.

Fluorescence imaging and quantification

The green autofluorescence signals achieved by the expression of GFP in HEK-293 and tsA201 cells were imaged on a VWR Trinocular Inverted Microscope and captured using moticam pro S5 lite camera and a C-mount adapter 0.65X. For each 35 mm cell culture dish, we took three fluorescence images from three randomly chosen locations under a $10\times$ objective lens. For measuring the fluorescence intensity, each image was uploaded to ImageJ 1.52a, and the intensity was measured by drawing a rectangle of the same size on 5 different random locations on each image. The intensity of each rectangle was calculated by the average of its pixel scores with each pixel having a value ranging from 0 to 255 on an 8-bit digital scale.⁵⁸ Since the intensity of the fluorescence signal in Figure 6 was low, we also calculated the percentage of fluorescent cells by splitting the cells into new dishes and culturing them in a 37°C incubator for only 1 h to be able to quantify cells before they make aggregates. We then quantified the number of fluorescent cells and divided them by the total number of cells in the corresponding light images.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed using GraphPad Prism 9 (GraphPad Software). Data were first tested for normal distribution using D'Agostino-Pearson Test. Unpaired two-tailed Student's t test and Mann-Whitney U test were used to compare two sets of normally and non-normally distributed data, respectively. One-way ANOVA was used for the comparison between any three sets of data. Tukey's post hoc test for multiple comparisons was used for normally distributed groups. Data were presented with error bars indicating the standard error of the mean (SEM). If at least one group showed a nonnormal distribution, one-way ANOVA on ranks followed by Dunn's post hoc test for multiple comparisons was used. Some data were presented as boxes extending from the 25th to the 75th percentile of each group's distribution of values with vertical lines and + signs denoting median and mean values, respectively. To compare the GFP fluorescence signal intensity, two-way ANOVA followed by Bonferroni's post hoc test was used with the compound treatment/non-treatment and the three randomly imaged areas of each transduced cell culture dish as the two factors. The significance was defined using a threshold of $p = 0.05$ throughout the study. The names of statistical tests are described in the figure legends. Sample sizes are described in the figures and figure legends.